

Carter Perez

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WORK EXPERIENCE

Cyber Skyline

Software Engineer

May. 2025 – Present

Remote

- Backend engineer for the CISA President's Cup CTF Competition, developed core competition systems including scoring engine, real-time notifications, support ticketing, and team/event management - #1 contributor with 56K+ lines.
- Engineered WebSocket notification system coordinating multi-instance Flask deployments via Redis pub/sub, implementing per-instance user session tracking to route messages across horizontally scaled containers without shared Socket.IO state
- Developed composable decorator middleware for authentication, resource loading, and error handling - standardizing security patterns and eliminating boilerplate across 60+ endpoints

Sealing Technologies

Integration Tech II

May. 2024 – May. 2025

Columbia, MD

- Managed and optimized QA workflows for custom server builds, aligning with strict timelines and regulatory standards..
- Upgraded and maintained computer hardware across the facility, ensuring optimal performance and seamless integration with custom server builds.
- Played a key role in consulting on the assembly and logistics of \$40m worth of servers, network switches, and Cyber kits.
- Authored multiple blogs for the company website detailing efficient quality assurance processes for high-stakes assembly projects.
- Developed SOPs to standardize assembly procedures across teams and streamline onboarding.

Jimmy Johns

General Manager

Dec. 2022 – June. 2024

Severna Park, MD

- Supervised 10+ staff members, optimizing task delegation and daily procedures for smooth restaurant operations.
- Diagnosed and repaired POS systems and networks across five stores, maintaining service reliability for the franchise.
- Optimized workflows and processes to eliminate excess labor costs and COGS, resulting in smoother operations.
- Reduced delivery times by 8 minutes weekly, securing recognition as the top-performing GM within a 5-store franchise.
- Created prep and cleaning checklists to improve back-of-house efficiency and accountability.

CERTIFICATIONS

CompTIA A+, Network+, Security+, CySA+, PenTest+, CASP+ | Python Institute PCEP

EDUCATION

University of Maryland Global Campus

MS, Cyber Security

May, 2027

Largo, MD

SKILLS

- **Technical Engineering & Development:** Splunk, AWS, Cloudflare, GCP, Docker, GitHub Actions, Linux, Python, Javascript, Typescript HTML, CSS, Shell, Regex, Scripting, Nginx, Apache, Git, Rust, C, Java, Swift, SolidJS, React, MongoDB, PostgreSQL, MySQL
- **Governance, Collaboration & Strategy:** Team building, Peer Mentoring, Certification Training, Customer Satisfaction, MITRE ATT&CK, ITIL, ISO 27001, COBIT, NIST 800-53, Cyber Kill Chains, CRM systems.

PROJECTS

- **Title:** [CertGames.com](https://certgames.com) | **iOS App:** <https://apps.apple.com/us/app/comptia-cert-games-practice/id6743811522>
- **GitHub:** github.com/CarterPerez-dev/CertGames-Core
- Developed full-stack cybersecurity learning platform and iOS app with 600+ active users, featuring gamification engine with XP/achievements, real-time leaderboards, virtual shop economy, and 15,000+ practice questions
- Established comprehensive testing pyramid with 5,250+ automated tests across unit, integration, and E2E layers achieving 96% code coverage and preventing 95% of regressions through custom pytest fixtures and CI/CD integration
- Implemented multi-layered security system with pattern detection, behavioral tracking, request validation, JWT token/IP blocklisting, and automated threat intelligence reducing attack surface by 90%
- **Title:** SolidJS + Rust Performance Showcase | **Website:** [SolidRust.com](https://solidrust.com)
- Built high-performance Rust backend with Axum framework featuring parallel fractal computation engine (Rayon), GitHub API integration, real-time performance monitoring with WebSocket support, and comprehensive system metrics collection across CPU, memory, disk, and application-level KPIs
- Developed reactive SolidJS frontend with TypeScript showcasing real-time performance dashboards, interactive fractal visualization canvas, Web Vitals tracking, and live system monitoring with <16ms render times and fine-grained reactivity eliminating virtual DOM overhead
- **Title:** FastAPI-Unified | **PyPi:** <https://pypi.org/project/fastapi-unified/0.1.1/>
- Built Flask/FastAPI decorator system consolidating scattered API concerns (validation, auth, rate limiting, audit) into single @unified decorator, reducing boilerplate by 80% and enabling declarative endpoint configuration with automatic dependency injection and response serialization
- Implemented JWT authentication system with scope-based permissions, optional/required auth modes, token blacklisting via JTI, and smart dependency injection; integrated Pydantic v2 schemas with custom validators for automatic request/response validation and ORM serialization
- **Title:** Semantic Media Searching | **Website:** <https://vuematics.com>
- Architected full-stack semantic media search platform using FastAPI/React enabling natural language queries over personal photo/video libraries; integrated Google Gemini 2.0 Flash for multimodal content analysis and OpenAI text-embedding-3-small (1536-dim) for vector embeddings, achieving sub-200ms search latency via PostgreSQL pgvector with cosine similarity and IVFFlat indexing
- Engineered async background processing pipeline handling image/video uploads with PIL/OpenCV frame extraction, automated thumbnail generation, and JWT-based authentication with refresh token rotation; implemented Redis caching, Slowapi rate limiting (100 req/min), and production Docker/Nginx deployment infrastructure
- **Title:** Flask-Honeypot | **PyPi:** <https://pypi.org/project/flask-honeypot/0.1.1/>
- Sophisticated cybersecurity deception system Python package that redirects 500+ common attack vectors to 20+ realistic-looking fake admin dashboards and portals
- Records all attacker interactions collecting 30+ data points including geolocation and behavioral patterns with built-in admin dashboard.

